AYMANE AIT DADS

Machine Learning Intern | Sensor Data Processing · Embedded Systems · Prototype Integration

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline for a 30B-parameter model, demonstrating full ownership of data processing, model training, and experimental documentation. At Orange Maroc, built machine learning models on 100K+ network sensor records, reducing manual data analysis time by 60% and delivering actionable outputs directly to an engineering optimization team. Proficient in Python, PyTorch, and Scikit-learn for data collection, processing, and analysis workflows, with hands-on experience building and documenting reproducible ML pipelines for experimental and sensor-based data. Aims to contribute machine learning model development and hardware-software integration skills to prototype systems supporting assistive and inclusive technology at the Center for Disability Services.

CORE COMPETENCIES

Machine Learning Models | Sensor-Based Data Analysis | Data Collection & Processing | Software Development | Embedded Systems | Hardware-Software Integration | Prototype Systems | Experiment Documentation

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed and implemented **Random Forest and K-Means machine learning models** on 100K+ fiber-optic network sensor records covering upload speed, latency, and jitter to classify network quality across thousands of nodes.
- Built an end-to-end data processing and analysis workflow that mapped customer performance profiles to **5 commercial service tiers**, with pipeline output adopted directly by the network optimization engineering team.
- Reduced manual data analysis time by **60%** through automated feature engineering and systematic model evaluation, enabling faster decision-making across the network operations workflow.
- Delivered stakeholder presentations translating model outputs into **maintenance recommendations**, documenting results and experimental findings clearly for both technical and non-technical audiences.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge — *Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in an ongoing challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Ran **controlled ablation experiments** across learning-rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data generation.

PyTorch · Unsloth · PEFT · TRL · Hugging Face Transformers · vLLM

Anomalous Sound Detection - Industrial Equipment — *EURECOM Course Project · 2025 · AUC 0.80*

- Designed an unsupervised fault-detection system for slide-rail industrial equipment by implementing a **Transformer autoencoder on Mel spectrograms** with SpecAugment, achieving AUC 0.80 for predictive maintenance without labeled anomaly data.
- Built the full data collection, processing, and model evaluation pipeline for **sensor-based audio signals**, documenting all experimental configurations and results to support reproducible prototype development.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | TensorFlow | Hugging Face Transformers | Unsloth | PEFT | TRL

Data Processing & Analysis: Python | Pandas | NumPy | feature engineering | SQL | Bash | Power BI

MLOps & Experimentation: Git | Linux | FP16/BF16 mixed precision | ablation studies | CUDA | Docker | REST APIs